



Syllabus: Matrices, determinants, System of linear equations, Eigenvalues and eigenvectors, LU decomposition.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	2	1	1	2	1	0	2	1	0	1	0
2 Marks Count	0	1	1	1	1	1	0	2	1	1	0
Total Marks	2	3	3	4	3	2	2	5	2	3	2

Welcome to the "Engineering Mathematics: Linear Algebra" chapter of your GATE Computer Science preparation. This section serves as a comprehensive, exam-focused reference to help you master the fundamental concepts and problem-solving techniques essential for success.

Subject Overview

Linear Algebra is the branch of mathematics concerning vector spaces and linear mappings between such spaces. It is a cornerstone of modern computing, underpinning areas like machine learning (e.g., PCA, SVMs, neural networks), computer graphics, data science, algorithm design, and optimization. For GATE CS, Linear Algebra typically contributes a significant portion of the Engineering Mathematics section, often carrying 3-5 marks out of the total 8-10 marks allocated to mathematics. Questions can range from direct computational problems (e.g., finding determinants, eigenvalues, ranks) to conceptual understanding (e.g., properties of vector spaces, subspaces, consistency of systems of equations) and application-based scenarios. Proficiency in this subject is not just about scoring marks but also about building a strong mathematical foundation for advanced computer science topics.

Topic-wise Key Concepts

Cartesian Coordinates

Cartesian coordinates provide a way to uniquely specify the position of a point in a space using numerical values. In 2D, a point is defined by its distance from two perpendicular axes (x, y), and in 3D, by its distance from three mutually perpendicular axes (x, y, z).

- **Definition:** A system that uses an ordered set of numbers (coordinates) to specify the location of a point in a space. For a 2D plane, it's (x, y) ; for 3D space, it's (x, y, z) .
- **Distance Formula (2D):** The distance between two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **Distance Formula (3D):** The distance between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

- **Midpoint Formula (2D):** The midpoint M of a line segment connecting $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Properties:**
 - Allows geometric problems to be translated into algebraic ones.
 - Forms the basis for vector representation and operations.
- **Common Pitfalls:**
 - Confusing x and y coordinates, especially when dealing with transformations.
 - Incorrectly applying distance or midpoint formulas.
- **Problem-Solving Techniques:**
 - Visualize the points and vectors in the coordinate system.
 - Break down complex geometric problems into simpler algebraic calculations.

Determinant

The determinant is a scalar value that can be computed from the elements of a square matrix. It provides crucial information about the matrix, such as its invertibility and the volume scaling factor of the linear transformation it represents.

• **Definition:** A scalar value associated with a square matrix A , denoted as $\det(A)$ or $|A|$.

• **Formulas:**

1. **For a 2×2 matrix:** If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $|A| = ad - bc$.

2. **For a 3×3 matrix (Sarrus' Rule):** If $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, then

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg). \text{ This can also be written as } a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

3. **General $n \times n$ matrix (Laplace Expansion/Cofactor Expansion):** $|A| = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$ (along row i) or $|A| = \sum_{i=1}^n (-1)^{i+j} a_{ij} M_{ij}$ (along column j) where M_{ij} is the minor (determinant of the submatrix obtained by deleting row i and column j).

• **Properties:**

- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A) \det(B)$
- $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix A and scalar k .
- If a matrix has a row or column of zeros, its determinant is 0.
- If a matrix has two identical rows or columns, its determinant is 0.
- If a matrix is triangular (upper or lower), its determinant is the product of its diagonal elements.
- Swapping two rows/columns changes the sign of the determinant.
- Adding a multiple of one row/column to another row/column does not change the determinant.
- A square matrix A is invertible if and only if $\det(A) \neq 0$.
- $\det(A^{-1}) = \frac{1}{\det(A)}$

• **Common Pitfalls:**

- Sign errors in cofactor expansion.
- Incorrectly applying properties, especially for scalar multiplication $\det(kA)$.
- Assuming $\det(A + B) = \det(A) + \det(B)$ (which is generally false).

• **Problem-Solving Techniques:**

- For larger matrices, use row/column operations to create zeros and simplify the calculation before applying cofactor expansion (e.g., make it triangular).
- Exploit properties (e.g., identical rows/columns, triangular form) to quickly find the determinant.
- Choose the row or column with the most zeros for cofactor expansion.

Eigen Value

Eigenvalues are special scalar values associated with a square matrix that characterize its fundamental properties. When a linear transformation is applied to an eigenvector, the eigenvector's direction remains unchanged, only its magnitude is scaled by the corresponding eigenvalue.

• **Definition:** For a square matrix A , a scalar λ is an eigenvalue if there exists a non-zero vector $\mathbf{v}$ (eigenvector) such that $A\mathbf{v} = \lambda\mathbf{v}$.

• **Formulas/Theorems:**

1. **Characteristic Equation:** To find eigenvalues, solve the equation $\det(A - \lambda I) = 0$, where I is the identity matrix of the same dimension as A .
2. **Cayley-Hamilton Theorem:** Every square matrix satisfies its own characteristic equation. If $P(\lambda) = \det(A - \lambda I)$ is the characteristic polynomial, then $P(A) = 0$. This is useful for finding powers of a matrix or its inverse.
3. **Sum of Eigenvalues:** The sum of the eigenvalues of a matrix A is equal to its trace (sum of diagonal elements): $\sum \lambda_i = \text{trace}(A)$.
4. **Product of Eigenvalues:** The product of the eigenvalues of a matrix A is equal to its determinant: $\prod \lambda_i = \det(A)$.

• **Properties:**

- If λ is an eigenvalue of A , then λ^k is an eigenvalue of A^k .
- If λ is an eigenvalue of A , then $1/\lambda$ is an eigenvalue of A^{-1} (if A is invertible).

- If λ is an eigenvalue of A , then $k\lambda$ is an eigenvalue of kA .
- The eigenvalues of a symmetric matrix are always real.
- Eigenvectors corresponding to distinct eigenvalues are linearly independent.
- For a real symmetric matrix, eigenvectors corresponding to distinct eigenvalues are orthogonal.
- The eigenvalues of a triangular matrix (upper or lower) are its diagonal entries.
- A matrix A is singular (non-invertible) if and only if 0 is an eigenvalue of A .
- **Common Pitfalls:**
 - Algebraic errors when solving the characteristic equation.
 - Forgetting to check for non-zero eigenvectors.
 - Confusing algebraic and geometric multiplicity.
- **Problem-Solving Techniques:**
 - Always start by setting up the characteristic equation $\det(A - \lambda I) = 0$.
 - For 2×2 matrices, use $\lambda^2 - \text{trace}(A)\lambda + \det(A) = 0$.
 - Use the sum and product of eigenvalues properties to verify your calculated eigenvalues or to find a missing eigenvalue.
 - For eigenvectors, substitute each eigenvalue back into $(A - \lambda I)\mathbf{v} = \mathbf{0}$ and solve the resulting system of equations.

Gaussian Elimination

Gaussian elimination is an algorithm used to solve systems of linear equations, find the rank of a matrix, compute the inverse of a matrix, and calculate determinants. It transforms a matrix into row echelon form using elementary row operations.

- **Definition:** A systematic procedure to transform a matrix into row echelon form (or reduced row echelon form) using elementary row operations.
- **Elementary Row Operations:**
 1. Swapping two rows ($R_i \leftrightarrow R_j$).
 2. Multiplying a row by a non-zero scalar ($kR_i \rightarrow R_i$).
 3. Adding a multiple of one row to another row ($R_i + kR_j \rightarrow R_i$).
- **Row Echelon Form (REF):** A matrix is in REF if:
 - All non-zero rows are above any rows of all zeros.
 - The leading entry (pivot) of each non-zero row is in a column to the right of the leading entry of the row above it.
 - All entries in a column below a leading entry are zeros.
- **Reduced Row Echelon Form (RREF):** A matrix is in RREF if it is in REF and:
 - The leading entry in each non-zero row is 1 (called a leading 1).
 - Each column containing a leading 1 has zeros everywhere else.
- **Applications:**
 - Solving systems of linear equations.
 - Finding the rank of a matrix.
 - Computing the inverse of a matrix (by augmenting $[A|I]$ and reducing to $[I|A^{-1}]$).
 - Calculating determinants (by keeping track of row operations).
- **Common Pitfalls:**
 - Arithmetic errors during row operations.
 - Forgetting to apply operations to the augmented part of the matrix when solving systems or finding inverses.
 - Incorrectly identifying pivot elements.
- **Problem-Solving Techniques:**
 - Always aim to create zeros below the pivot elements first.
 - Work column by column from left to right.
 - Use fractions carefully; sometimes it's better to multiply a row to avoid fractions temporarily.
 - For solving systems, use back-substitution after reaching REF.

LU Decomposition

LU decomposition factors a square matrix A into the product of a lower triangular matrix L and an upper triangular matrix U . This factorization is particularly useful for efficiently solving multiple systems of linear equations with the same coefficient matrix, as it simplifies the process into two triangular systems.

- **Definition:** A factorization of a square matrix A into the product of a lower triangular matrix L (with 1s on the diagonal) and an upper triangular matrix U , such that $A = LU$.
- **Process:**

1. Use Gaussian elimination to transform A into an upper triangular matrix U .
 2. The lower triangular matrix L is constructed using the multipliers used in the Gaussian elimination process. If $R_i - kR_j \rightarrow R_i$ was used, then $L_{ij} = k$. The diagonal elements of L are 1s.
 3. If row swaps are needed, a permutation matrix P is introduced, such that $PA = LU$.
- **Solving $A\mathbf{x} = \mathbf{b}$ using LU Decomposition:**
 1. Substitute $A = LU$: $LU\mathbf{x} = \mathbf{b}$.
 2. Let $U\mathbf{x} = \mathbf{y}$.
 3. Solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ using forward substitution (since L is lower triangular).
 4. Solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ using backward substitution (since U is upper triangular).
 - **Properties:**
 - Not all matrices have an LU decomposition without pivoting (row swaps).
 - If A is non-singular and all its leading principal minors are non-zero, then A has a unique LU decomposition.
 - **Common Pitfalls:**
 - Incorrectly forming L from the multipliers.
 - Errors in forward or backward substitution.
 - Forgetting the permutation matrix P if row swaps are involved.
 - **Problem-Solving Techniques:**
 - Carefully track the multipliers used in Gaussian elimination to construct L .
 - Practice forward and backward substitution for efficiency.
 - Recognize when pivoting is necessary (e.g., if a pivot element becomes zero).

Matrix

A matrix is a rectangular array of numbers, symbols, or expressions arranged in rows and columns. Matrices are fundamental to linear algebra, representing linear transformations, systems of equations, and data structures.

- **Definition:** A rectangular array of numbers (or functions) arranged in m rows and n columns. Denoted as $A_{m \times n}$.
- **Types of Matrices:**
 - **Square Matrix:** $m = n$.
 - **Row Matrix:** $m = 1$.
 - **Column Matrix (Vector):** $n = 1$.
 - **Identity Matrix (I):** Square matrix with 1s on the main diagonal and 0s elsewhere. $AI = IA = A$.
 - **Zero Matrix (0):** All elements are 0. $A + 0 = A$.
 - **Diagonal Matrix:** Square matrix where all non-diagonal elements are 0.
 - **Scalar Matrix:** A diagonal matrix where all diagonal elements are equal.
 - **Symmetric Matrix:** $A = A^T$.
 - **Skew-Symmetric Matrix:** $A = -A^T$. Diagonal elements are 0.
 - **Orthogonal Matrix:** $AA^T = A^T A = I$, which implies $A^{-1} = A^T$. Determinant is ± 1 .
 - **Invertible/Non-singular Matrix:** A square matrix A for which there exists a matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$. $\det(A) \neq 0$.
 - **Singular Matrix:** A square matrix A for which $\det(A) = 0$. It does not have an inverse.
- **Operations:**
 1. **Addition/Subtraction:** Element-wise, only for matrices of the same dimensions. $(A + B)_{ij} = A_{ij} + B_{ij}$.
 2. **Scalar Multiplication:** Multiply each element by the scalar. $(kA)_{ij} = kA_{ij}$.
 3. **Matrix Multiplication:** $(AB)_{ij} = \sum_{k=1}^p A_{ik}B_{kj}$. Requires number of columns of A to equal number of rows of B . Resulting matrix has dimensions $m \times q$ if A is $m \times p$ and B is $p \times q$.
 4. **Transpose (A^T):** Rows become columns and columns become rows. $(A^T)_{ij} = A_{ji}$.
 5. **Inverse (A^{-1}):** For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. For general $n \times n$, $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$, where $\text{adj}(A)$ is the adjugate matrix (transpose of the cofactor matrix).
- **Properties:**
 - Matrix addition is commutative: $A + B = B + A$.
 - Matrix multiplication is associative: $(AB)C = A(BC)$.
 - Matrix multiplication is generally NOT commutative: $AB \neq BA$.
 - $(A + B)^T = A^T + B^T$.
 - $(AB)^T = B^T A^T$.
 - $(A^{-1})^{-1} = A$.

- $(AB)^{-1} = B^{-1}A^{-1}$.
- $(A^T)^{-1} = (A^{-1})^T$.
- **Common Pitfalls:**
 - Incorrect dimensions for addition or multiplication.
 - Assuming commutativity for matrix multiplication.
 - Errors in calculating inverses, especially for larger matrices.
- **Problem-Solving Techniques:**
 - Always check dimensions before performing operations.
 - For matrix multiplication, visualize the dot products of rows and columns.
 - For inverses, use the adjugate formula for 2×2 and Gaussian elimination for larger matrices.

Multiplicity

Multiplicity refers to how many times an eigenvalue appears as a root of the characteristic polynomial (algebraic multiplicity) and the dimension of its corresponding eigenspace (geometric multiplicity). These concepts are crucial for understanding diagonalizability of a matrix.

- **Definition:**
 - **Algebraic Multiplicity (AM):** The number of times an eigenvalue λ appears as a root of the characteristic polynomial $\det(A - \lambda I) = 0$.
 - **Geometric Multiplicity (GM):** The dimension of the eigenspace corresponding to λ , which is the number of linearly independent eigenvectors associated with λ . It is given by $n - \text{rank}(A - \lambda I)$, where n is the dimension of the matrix.
- **Properties/Theorems:**
 - For any eigenvalue λ , $1 \leq \text{GM}(\lambda) \leq \text{AM}(\lambda)$.
 - A matrix A is diagonalizable if and only if for every eigenvalue λ , $\text{AM}(\lambda) = \text{GM}(\lambda)$.
 - If all eigenvalues are distinct, then $\text{AM}(\lambda) = \text{GM}(\lambda) = 1$ for all λ , and the matrix is diagonalizable.
- **Common Pitfalls:**
 - Confusing AM and GM.
 - Incorrectly calculating the rank of $(A - \lambda I)$.
 - Assuming diagonalizability based only on AM without checking GM.
- **Problem-Solving Techniques:**
 - First, find all eigenvalues and their algebraic multiplicities by solving $\det(A - \lambda I) = 0$.
 - For each eigenvalue, calculate the rank of $(A - \lambda I)$ to find its geometric multiplicity.
 - Compare AM and GM for each eigenvalue to determine diagonalizability.

Orthonormality

Orthonormality describes a set of vectors that are mutually orthogonal (perpendicular) and each has a unit length (normalized). Orthonormal bases simplify many calculations in linear algebra, particularly in projections and coordinate transformations.

- **Definition:**
 - **Orthogonal Vectors:** Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if their dot product is zero: $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = 0$.
 - **Normalized Vector:** A vector $\mathbf{u}$ is normalized if its length (magnitude) is 1: $\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = 1$.
 - **Orthonormal Set:** A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is orthonormal if they are mutually orthogonal and each vector is normalized. That is, $\mathbf{v}_i \cdot \mathbf{v}_j = \delta_{ij}$ (Kronecker delta), where $\delta_{ij} = 1$ if $i = j$ and 0 if $i \neq j$.
- **Gram-Schmidt Orthonormalization Process:** A procedure to transform a set of linearly independent vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ into an orthonormal set $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$.
 1. $\mathbf{v}_1 = \mathbf{x}_1$
 2. $\mathbf{v}_2 = \mathbf{x}_2 - \text{proj}_{\mathbf{v}_1} \mathbf{x}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \mathbf{v}_1$
 3. $\mathbf{v}_3 = \mathbf{x}_3 - \text{proj}_{\mathbf{v}_1} \mathbf{x}_3 - \text{proj}_{\mathbf{v}_2} \mathbf{x}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\|\mathbf{v}_2\|^2} \mathbf{v}_2$
 4. Continue for all vectors.
 5. Normalize each $\mathbf{v}_i$: $\mathbf{u}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}$.
- **Properties:**
 - An orthonormal set of vectors is always linearly independent.
 - If Q is an orthogonal matrix (columns form an orthonormal set), then $Q^T Q = I$, and $Q^{-1} = Q^T$.
 - For an orthogonal matrix Q , $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ (preserves length).
 - For an orthogonal matrix Q , $(Q\mathbf{x}) \cdot (Q\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ (preserves dot product and angle).

- **Common Pitfalls:**

- Arithmetic errors in Gram-Schmidt, especially with projections.
- Forgetting to normalize vectors after making them orthogonal.
- Confusing orthogonal with orthonormal.

- **Problem-Solving Techniques:**

- Systematically apply the Gram-Schmidt process, calculating one orthogonal vector at a time.
- Double-check dot products to ensure orthogonality.
- Always normalize the final set of orthogonal vectors.

Rank of Matrix

The rank of a matrix is a fundamental property that measures the "linear independence" present in its rows and columns. It indicates the dimension of the vector space spanned by its rows or columns, and is crucial for understanding the solvability of linear systems.

- **Definition:** The maximum number of linearly independent row vectors (row rank) or column vectors (column rank) in a matrix. The row rank is always equal to the column rank.

- **Calculation Methods:**

1. **Using Row Echelon Form:** The rank of a matrix is the number of non-zero rows in its row echelon form.
2. **Using Determinants (Minors):** The rank of a matrix A is the largest integer r such that there exists an $r \times r$ submatrix of A with a non-zero determinant.

- **Properties/Theorems:**

- For an $m \times n$ matrix A , $\text{rank}(A) \leq \min(m, n)$.
- $\text{rank}(A) = 0$ if and only if A is a zero matrix.
- $\text{rank}(A) = \text{rank}(A^T)$.
- $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.
- **Rank-Nullity Theorem:** For an $m \times n$ matrix A , $\text{rank}(A) + \text{nullity}(A) = n$, where $\text{nullity}(A)$ is the dimension of the null space (kernel) of A .
- A square matrix A of size $n \times n$ is invertible if and only if $\text{rank}(A) = n$.
- For a system $A\mathbf{x} = \mathbf{b}$:
 - Consistent if $\text{rank}(A) = \text{rank}([A|\mathbf{b}])$.
 - Unique solution if $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) = n$ (number of variables).
 - Infinitely many solutions if $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) < n$.
 - Inconsistent if $\text{rank}(A) \neq \text{rank}([A|\mathbf{b}])$.

- **Common Pitfalls:**

- Errors in Gaussian elimination leading to incorrect row echelon form.
- Misinterpreting the number of non-zero rows.
- Not considering the augmented matrix for system consistency.

- **Problem-Solving Techniques:**

- The most reliable method is to reduce the matrix to row echelon form using Gaussian elimination and count the number of non-zero rows.
- For smaller matrices, checking determinants of submatrices can be faster.
- Use the Rank-Nullity Theorem to find nullity if rank is known, or vice versa.

Singular Value Decomposition (SVD)

Singular Value Decomposition is a powerful matrix factorization technique that decomposes any $m \times n$ matrix A into three matrices: U , Σ , and V^T . It generalizes the concept of eigenvalues and eigenvectors to non-square matrices and has broad applications in data compression, noise reduction, and recommender systems.

- **Definition:** Any $m \times n$ matrix A can be factored as $A = U\Sigma V^T$, where:

- U is an $m \times m$ orthogonal matrix whose columns are the left singular vectors of A .
- Σ is an $m \times n$ diagonal matrix with non-negative real numbers on the diagonal, called singular values (σ_i), arranged in decreasing order. The non-zero singular values are $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, where $r = \text{rank}(A)$.
- V is an $n \times n$ orthogonal matrix whose columns are the right singular vectors of A . V^T is its transpose.

- **Relationship to Eigenvalues:**

- The singular values σ_i of A are the square roots of the eigenvalues of $A^T A$ (or AA^T).
- The columns of V are the eigenvectors of $A^T A$.
- The columns of U are the eigenvectors of AA^T .

- **Properties/Applications:**

- SVD exists for any matrix (square or rectangular).
- The number of non-zero singular values is equal to the rank of the matrix.
- Low-rank approximation: By keeping only the largest singular values and corresponding singular vectors, SVD can approximate a matrix with fewer components, useful for data compression.
- Pseudo-inverse: The SVD can be used to compute the Moore-Penrose pseudo-inverse of a matrix.
- Principal Component Analysis (PCA): SVD is closely related to PCA, where singular vectors correspond to principal components.

- **Common Pitfalls:**

- Confusing singular values with eigenvalues.
- Incorrectly calculating eigenvectors for $A^T A$ or AA^T .
- Not arranging singular values in decreasing order.

- **Problem-Solving Techniques:**

- Calculate $A^T A$ (or AA^T).
- Find the eigenvalues of $A^T A$. The square roots of these eigenvalues are the singular values σ_i .
- Find the eigenvectors of $A^T A$. These form the columns of V .
- Find the eigenvectors of AA^T . These form the columns of U .
- Alternatively, $U_i = \frac{1}{\sigma_i} A V_i$.

Statistics

While a broad field, within the context of linear algebra for GATE CS, statistics primarily involves concepts like mean, variance, covariance, and their matrix representations. These are crucial for understanding data analysis techniques like PCA, which heavily rely on linear algebra.

- **Definition (Linear Algebra Context):** Focuses on the mathematical tools (vectors, matrices) used to analyze and model data, particularly in multivariate statistics.

- **Key Concepts:**

1. **Mean:** For a vector $\mathbf{x} = [x_1, \dots, x_n]^T$, the mean is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.
2. **Variance:** A measure of how spread out the data is. For a vector $\mathbf{x}$, $\text{Var}(\mathbf{x}) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$.
3. **Covariance:** A measure of how two variables change together. For two vectors $\mathbf{x}$ and $\mathbf{y}$,
 $\text{Cov}(\mathbf{x}, \mathbf{y}) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$.
4. **Covariance Matrix:** For a dataset with p variables and n observations, the covariance matrix Σ is a $p \times p$ symmetric matrix where $\Sigma_{ij} = \text{Cov}(\mathbf{x}_i, \mathbf{x}_j)$ and $\Sigma_{ii} = \text{Var}(\mathbf{x}_i)$.

$$\Sigma = \frac{1}{n-1} X^T X$$

where X is the data matrix with columns centered (mean subtracted).

5. **Principal Component Analysis (PCA):** A dimensionality reduction technique that uses the eigenvectors of the covariance matrix (or SVD of the data matrix) to find orthogonal directions (principal components) of maximum variance in the data.

- **Properties:**

- Covariance matrix is always symmetric and positive semi-definite.
- Eigenvalues of the covariance matrix represent the variance along the principal components.
- Eigenvectors of the covariance matrix are the principal components.

- **Common Pitfalls:**

- Confusing variance with standard deviation.
- Incorrectly calculating covariance or covariance matrix.
- Not centering data before calculating covariance matrix for PCA.

- **Problem-Solving Techniques:**

- Understand the definitions of mean, variance, and covariance.
- For PCA, compute the covariance matrix, then find its eigenvalues and eigenvectors.
- Relate SVD to PCA: SVD of the centered data matrix directly gives principal components and singular values related to variance.

Subspace

A subspace is a subset of a vector space that itself satisfies the properties of a vector space. It must contain the zero vector, be closed under vector addition, and closed under scalar multiplication. Subspaces are fundamental building

blocks for understanding the structure of vector spaces.

- **Definition:** A subset W of a vector space V is a subspace of V if W is itself a vector space under the operations defined on V .
- **Conditions for a Subspace:** A non-empty subset W of a vector space V is a subspace if and only if:
 1. The zero vector of V is in W ($\mathbf{0} \in W$).
 2. W is closed under vector addition: If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$.
 3. W is closed under scalar multiplication: If $\mathbf{u} \in W$ and c is any scalar, then $c\mathbf{u} \in W$.
- **Important Subspaces Associated with a Matrix A :**
 - **Column Space ($\text{Col}(A)$):** The span of the column vectors of A . It is a subspace of $\mathbb{R}^m$ (if A is $m \times n$).
 $\dim(\text{Col}(A)) = \text{rank}(A)$.
 - **Row Space ($\text{Row}(A)$):** The span of the row vectors of A . It is a subspace of $\mathbb{R}^n$. $\dim(\text{Row}(A)) = \text{rank}(A)$.
 - **Null Space ($\text{Null}(A)$ or $\text{Ker}(A)$):** The set of all vectors $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$. It is a subspace of $\mathbb{R}^n$.
 $\dim(\text{Null}(A)) = \text{nullity}(A)$.
 - **Left Null Space ($\text{Null}(A^T)$):** The set of all vectors $\mathbf{y}$ such that $A^T\mathbf{y} = \mathbf{0}$. It is a subspace of $\mathbb{R}^m$.
- **Properties:**
 - The intersection of two subspaces is always a subspace.
 - The union of two subspaces is generally not a subspace.
 - The sum of two subspaces $W_1 + W_2 = \{\mathbf{w}_1 + \mathbf{w}_2 \mid \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2\}$ is a subspace.
- **Common Pitfalls:**
 - Forgetting to check all three conditions for a subspace (especially the zero vector).
 - Confusing the column space with the null space.
 - Assuming the union of subspaces is a subspace.
- **Problem-Solving Techniques:**
 - To check if a set is a subspace, verify the three conditions.
 - To find a basis for the column space, identify the pivot columns of the original matrix after reducing to REF.
 - To find a basis for the null space, solve $A\mathbf{x} = \mathbf{0}$ and express the solution in terms of free variables.
 - To find a basis for the row space, use the non-zero rows of the REF of the matrix.

System of Equations

A system of linear equations is a collection of one or more linear equations involving the same set of variables. Linear algebra provides powerful tools to determine the existence and nature of solutions (unique, infinite, or no solution) for such systems.

- **Definition:** A set of equations of the form $a_{11}x_1 + \dots + a_{1n}x_n = b_1, \dots, a_{m1}x_1 + \dots + a_{mn}x_n = b_m$. This can be written in matrix form as $A\mathbf{x} = \mathbf{b}$, where A is the coefficient matrix, $\mathbf{x}$ is the vector of variables, and $\mathbf{b}$ is the constant vector.
- **Types of Systems:**
 - **Homogeneous System:** $A\mathbf{x} = \mathbf{0}$. Always consistent (has at least the trivial solution $\mathbf{x} = \mathbf{0}$).
 - **Non-homogeneous System:** $A\mathbf{x} = \mathbf{b}$ where $\mathbf{b} \neq \mathbf{0}$.
- **Consistency and Number of Solutions (using Rank):**
 1. **Consistent System:** A system has at least one solution if $\text{rank}(A) = \text{rank}([A|\mathbf{b}])$.
 - **Unique Solution:** If $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) = n$ (number of variables).
 - **Infinitely Many Solutions:** If $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) < n$. The number of free variables is $n - \text{rank}(A)$.
 2. **Inconsistent System:** A system has no solution if $\text{rank}(A) \neq \text{rank}([A|\mathbf{b}])$.
- **Solution Methods:**
 - **Gaussian Elimination/Row Reduction:** Transform the augmented matrix $[A|\mathbf{b}]$ into row echelon form to solve by back-substitution.
 - **Matrix Inverse Method:** If A is square and invertible, $\mathbf{x} = A^{-1}\mathbf{b}$. (Only for unique solutions).
 - **Cramer's Rule:** For a square system with a unique solution, $x_i = \frac{\det(A_i)}{\det(A)}$, where A_i is the matrix formed by replacing the i -th column of A with $\mathbf{b}$. (Computationally expensive for larger systems).
- **Properties of Homogeneous Systems:**
 - Always has the trivial solution $\mathbf{x} = \mathbf{0}$.
 - Has non-trivial solutions if and only if $\det(A) = 0$ (for square A) or $\text{rank}(A) < n$.
 - The set of all solutions forms the null space of A .
- **Common Pitfalls:**
 - Arithmetic errors during row operations.
 - Incorrectly determining consistency or number of solutions.

- Applying Cramer's rule or inverse method when not applicable (e.g., non-square matrix, singular matrix).
- **Problem-Solving Techniques:**
 - Always form the augmented matrix $[A|\mathbf{b}]$.
 - Use Gaussian elimination to reduce the augmented matrix to row echelon form.
 - Analyze the rank of A and $[A|\mathbf{b}]$ to determine consistency and number of solutions.
 - For homogeneous systems, look for non-trivial solutions if $\det(A) = 0$ or $\text{rank}(A) < n$.

Vector Space

A vector space is a fundamental algebraic structure consisting of a set of vectors, along with two operations: vector addition and scalar multiplication, which satisfy a set of ten axioms. It provides a generalized framework for working with vectors beyond simple geometric arrows.

- **Definition:** A non-empty set V of objects, called vectors, on which two operations are defined: vector addition (denoted by $+$) and scalar multiplication (denoted by juxtaposition), subject to ten axioms. The scalars are typically real numbers ($\mathbb{R}$) or complex numbers ($\mathbb{C}$).
- **Axioms of a Vector Space:** For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars c, d :
 1. $\mathbf{u} + \mathbf{v} \in V$ (Closure under addition)
 2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (Commutativity of addition)
 3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (Associativity of addition)
 4. There exists a zero vector $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$ (Additive identity)
 5. For each $\mathbf{u} \in V$, there exists an additive inverse $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ (Additive inverse)
 6. $c\mathbf{u} \in V$ (Closure under scalar multiplication)
 7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ (Distributivity of scalar over vector addition)
 8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ (Distributivity of scalar over scalar addition)
 9. $c(d\mathbf{u}) = (cd)\mathbf{u}$ (Associativity of scalar multiplication)
 10. $1\mathbf{u} = \mathbf{u}$ (Multiplicative identity)
- **Key Concepts:**
 - **Span:** The set of all possible linear combinations of a set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. Denoted as $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. It is always a subspace.
 - **Linear Independence:** A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent if the only solution to $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0}$ is $c_1 = \dots = c_k = 0$.
 - **Basis:** A set of vectors in a vector space V that is linearly independent and spans V . The number of vectors in a basis is unique and is called the dimension of the vector space.
 - **Dimension:** The number of vectors in any basis for a vector space V , denoted as $\dim(V)$.
- **Examples of Vector Spaces:**
 - $\mathbb{R}^n$ (n -dimensional real coordinate space)
 - The set of all $m \times n$ matrices.
 - The set of all polynomials of degree at most n .
 - The set of all continuous functions on an interval.
- **Common Pitfalls:**
 - Forgetting to check all axioms when verifying if a set is a vector space.
 - Confusing a vector space with a subspace.
 - Incorrectly determining linear independence or a basis.
- **Problem-Solving Techniques:**
 - To check if a set is a vector space, verify all ten axioms.
 - To check linear independence, set up the equation $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0}$ and solve the resulting system of linear equations.
 - To find a basis, identify a linearly independent set that spans the space. For $\mathbb{R}^n$, this often involves finding pivot columns of a matrix formed by the vectors.

Quick Formula Reference

- **Determinant:**
 - 2×2 : $|A| = ad - bc$ for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$
 - Properties: $|A^T| = |A|$, $|AB| = |A||B|$, $|kA| = k^n|A|$, $|A^{-1}| = 1/|A|$
 - Invertible iff $|A| \neq 0$
- **Matrix Operations:**
 - Matrix Multiplication: $(AB)_{ij} = \sum_k A_{ik}B_{kj}$

- Inverse 2×2 : $A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
- General Inverse: $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$
- Transpose Properties: $(A+B)^T = A^T + B^T$, $(AB)^T = B^T A^T$
- Inverse Properties: $(A^{-1})^{-1} = A$, $(AB)^{-1} = B^{-1} A^{-1}$
- **Eigenvalues & Eigenvectors:**
 - Characteristic Equation: $\det(A - \lambda I) = 0$
 - Sum of Eigenvalues: $\sum \lambda_i = \text{trace}(A)$
 - Product of Eigenvalues: $\prod \lambda_i = \det(A)$
 - Cayley-Hamilton Theorem: $P(A) = 0$ where $P(\lambda)$ is the characteristic polynomial.
 - Algebraic Multiplicity (AM): Number of times λ is a root.
 - Geometric Multiplicity (GM): $\dim(\text{Null}(A - \lambda I)) = n - \text{rank}(A - \lambda I)$
 - Diagonalizable if $\text{AM}(\lambda) = \text{GM}(\lambda)$ for all λ .
- **Rank of Matrix:**
 - Number of non-zero rows in REF.
 - $\text{rank}(A) \leq \min(m, n)$ for $m \times n$ matrix.
 - Rank-Nullity Theorem: $\text{rank}(A) + \text{nullity}(A) = n$ (number of columns).
 - System $A\mathbf{x} = \mathbf{b}$ consistent if $\text{rank}(A) = \text{rank}([A|\mathbf{b}])$.
 - Unique solution if $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) = n$.
 - Infinite solutions if $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) < n$.
 - No solution if $\text{rank}(A) \neq \text{rank}([A|\mathbf{b}])$.
- **LU Decomposition:**
 - $A = LU$ (or $PA = LU$ with permutation matrix P)
 - Solve $A\mathbf{x} = \mathbf{b}$ by $L\mathbf{y} = \mathbf{b}$ (forward substitution) then $U\mathbf{x} = \mathbf{y}$ (backward substitution).
- **Orthonormality:**
 - Orthogonal: $\mathbf{u} \cdot \mathbf{v} = 0$
 - Normalized: $\|\mathbf{u}\| = 1$
 - Orthonormal: $\mathbf{u}_i \cdot \mathbf{u}_j = \delta_{ij}$
 - Orthogonal Matrix Q : $Q^T Q = I \implies Q^{-1} = Q^T$.
- **Singular Value Decomposition (SVD):**
 - $A = U \Sigma V^T$
 - Singular values σ_i are $\sqrt{\lambda_i(A^T A)}$.
 - Columns of V are eigenvectors of $A^T A$.
 - Columns of U are eigenvectors of AA^T .
- **Subspace Conditions:**
 - Contains zero vector.
 - Closed under addition.
 - Closed under scalar multiplication.

Important Tips for GATE

1. **Master the Basics:** Ensure a strong understanding of fundamental definitions (e.g., matrix types, vector space axioms, linear independence). Many conceptual questions test these directly.
2. **Practice Gaussian Elimination:** This is a core algorithm. Practice it diligently for solving systems, finding rank, and computing inverses. Accuracy and speed are crucial here.
3. **Understand Eigenvalue/Eigenvector Properties:** Don't just memorize formulas. Understand *why* the sum of eigenvalues equals the trace and the product equals the determinant. These properties are often used as shortcuts or for verification.
4. **Pay Attention to Matrix Dimensions:** Always check if matrix operations (addition, multiplication) are valid based on dimensions. This prevents common errors, especially in matrix multiplication.
5. **Rank-Nullity Theorem is Your Friend:** This theorem is incredibly useful for quickly finding the nullity if the rank is known, or vice versa, and for analyzing the nature of solutions to linear systems.
6. **Beware of Calculation Errors:** Linear algebra problems often involve extensive calculations. Use the virtual calculator carefully and double-check intermediate steps. Simple arithmetic mistakes are a common reason for losing marks.
7. **Distinguish Between Similar Concepts:** Clearly differentiate between algebraic and geometric multiplicity, orthogonal and orthonormal vectors, and the various matrix subspaces (column space, null space, row space).
8. **Practice Previous Year Questions:** Solve as many GATE previous year questions as possible. This will familiarize

you with the typical question patterns, difficulty levels, and time constraints. Focus on understanding the solution approach rather than just getting the answer.

6.1

Cartesian Coordinates (1)

6.1.1 Cartesian Coordinates: GATE IT 2007 | Question: 80



Let $P_1, P_2, \dots, P_n$ be n points in the xy -plane such that no three of them are collinear. For every pair of points P_i and P_j , let L_{ij} be the line passing through them. Let L_{ab} be the line with the steepest gradient amongst all $\frac{n(n-1)}{2}$ lines.

Which one of the following properties should necessarily be satisfied ?

- A. P_a and P_b are adjacent to each other with respect to their x -coordinate
- B. Either P_a or P_b has the largest or the smallest y -coordinate among all the points
- C. The difference between x -coordinates P_a and P_b is minimum
- D. None of the above

gateit-2007 linear-algebra cartesian-coordinates

Answer key

6.2

Determinant (12)

Practice Tests: Test 1 (15Q) Test 2 (4Q)

6.2.1 Determinant: GATE CSE 1997 | Question: 1.3



The determinant of the matrix $\begin{bmatrix} 6 & -8 & 1 & 1 \\ 0 & 2 & 4 & 6 \\ 0 & 0 & 4 & 8 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

- A. 11
- B. -48
- C. 0
- D. -24

gate1997 linear-algebra normal determinant

Answer key

6.2.2 Determinant: GATE CSE 2000 | Question: 1.3



The determinant of the matrix

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 8 & 1 & 7 & 2 \\ 2 & 0 & 2 & 0 \\ 9 & 0 & 6 & 1 \end{bmatrix}$$

- A. 4
- B. 0
- C. 15
- D. 20

gatecse-2000 linear-algebra easy determinant

Answer key

6.2.3 Determinant: GATE CSE 2013 | Question: 3



Which one of the following does NOT equal

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} ?$$

A. $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$

C. $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$

gatecse-2013 linear-algebra normal determinant

Answer key

B. $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

D. $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

6.2.4 Determinant: GATE CSE 2014 | Set 2 | Question: 4



If the matrix A is such that

$$A = \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} [1 \ 9 \ 5]$$

then the determinant of A is equal to _____.

gatecse-2014-set2 linear-algebra numerical-answers easy determinant

Answer key

6.2.5 Determinant: GATE CSE 2019 | Question: 9



Let X be a square matrix. Consider the following two statements on X .

- I. X is invertible
- II. Determinant of X is non-zero

Which one of the following is TRUE?

- A. I implies II; II does not imply I
- B. II implies I; I does not imply II
- C. I does not imply II; II does not imply I
- D. I and II are equivalent statements

gatecse-2019 engineering-mathematics linear-algebra determinant one-mark

Answer key

6.2.6 Determinant: GATE CSE 2023 | Question: 8



Let

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$

Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B , respectively.

Which one of the options given below is TRUE?

- A. $\det(A) = \det(B)$

- B. $\det(B) = -\det(A)$
- C. $\det(A) = 0$
- D. $\det(AB) = \det(A) + \det(B)$

gatecse-2023 linear-algebra determinant one-mark easy

Answer key

6.2.7 Determinant: GATE CSE 2024 | Set 2 | Question: 37



Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- A. The determinant of B is the negative of the determinant of A
- B. If A is invertible, then B is also invertible
- C. If A is symmetric, then B is also symmetric
- D. If the trace of A is zero, then the trace of B is also zero

gatecse-2024-set2 linear-algebra multiple-selects matrix determinant two-marks

Answer key

6.2.8 Determinant: GATE CSE 2025 | Set 2 | Question: 4



Let L , M , and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$.

Which ONE of the following is the value of the determinant of $(M - N)$?

- A. 0
- B. 1
- C. 2
- D. 3

gatecse2025-set2 linear-algebra determinant easy one-mark

Answer key

6.2.9 Determinant: GATE CSE 2026 | Set 2 | Question: 52



The determinant of a 4×4 matrix A is 3. The value of the determinant of $2A$ is _____. (answer in integer)

gatecse-2026-set2 linear-algebra determinant numerical-answers two-marks

Answer key

6.2.10 Determinant: GATE DS&AI 2024 | Question: 25



Consider the 3×3 matrix $M = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 3 \\ 4 & 3 & 6 \end{bmatrix}$.

The determinant of $(M^2 + 12M)$ is _____.

gate-ds-ai-2024 numerical-answers matrix determinant linear-algebra easy one-mark

Answer key

6.2.11 Determinant: GATE IT 2004 | Question: 32



Let A be an $n \times n$ matrix of the following form.

$$A = \begin{bmatrix} 3 & 1 & 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 & \dots & 0 & 0 & 0 \\ \dots & & & & & & & & \\ \dots & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \dots & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 & 1 & 3 \end{bmatrix}_{n \times n}$$

What is the value of the determinant of A ?

- A. $\left(\frac{5+\sqrt{3}}{2}\right)^{n-1} \left(\frac{5\sqrt{3}+7}{2\sqrt{3}}\right) + \left(\frac{5-\sqrt{3}}{2}\right)^{n-1} \left(\frac{5\sqrt{3}-7}{2\sqrt{3}}\right)$
 B. $\left(\frac{7+\sqrt{5}}{2}\right)^{n-1} \left(\frac{7\sqrt{5}+3}{2\sqrt{5}}\right) + \left(\frac{7-\sqrt{5}}{2}\right)^{n-1} \left(\frac{7\sqrt{5}-3}{2\sqrt{5}}\right)$
 C. $\left(\frac{3+\sqrt{7}}{2}\right)^{n-1} \left(\frac{3\sqrt{7}+5}{2\sqrt{7}}\right) + \left(\frac{3-\sqrt{7}}{2}\right)^{n-1} \left(\frac{3\sqrt{7}-5}{2\sqrt{7}}\right)$
 D. $\left(\frac{3+\sqrt{5}}{2}\right)^{n-1} \left(\frac{3\sqrt{5}+7}{2\sqrt{5}}\right) + \left(\frac{3-\sqrt{5}}{2}\right)^{n-1} \left(\frac{3\sqrt{5}-7}{2\sqrt{5}}\right)$

gateit-2004 linear-algebra matrix normal determinant

Answer key

6.2.12 Determinant: GATE IT 2005 | Question: 3

The determinant of the matrix given below is

$$\begin{bmatrix} 0 & 1 & 0 & 2 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 1 & -2 & 0 & 1 \end{bmatrix}$$

- A. -1 B. 0 C. 1 D. 2

gateit-2005 linear-algebra normal determinant

Answer key

6.3 Eigen Value (33)

 **Practice Tests:** [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(13Q\)](#)

6.3.1 Eigen Value: GATE CSE 1993 | Question: 01.1

The eigen vector (s) of the matrix

$$\begin{bmatrix} 0 & 0 & \alpha \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \alpha \neq 0$$

is (are)

- A. $(0,0,\alpha)$ B. $(\alpha,0,0)$ C. $(0,0,1)$ D. $(0,\alpha,0)$

gate1993 eigen-value linear-algebra easy multiple-selects

Answer key

6.3.2 Eigen Value: GATE CSE 2002 | Question: 5a



Obtain the eigen values of the matrix

$$A = \begin{bmatrix} 1 & 2 & 34 & 49 \\ 0 & 2 & 43 & 94 \\ 0 & 0 & -2 & 104 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

gatecse-2002 linear-algebra eigen-value normal descriptive

Answer key

6.3.3 Eigen Value: GATE CSE 2005 | Question: 49



What are the eigenvalues of the following 2×2 matrix?

$$\begin{pmatrix} 2 & -1 \\ -4 & 5 \end{pmatrix}$$

- A. -1 and 1 B. 1 and 6 C. 2 and 5 D. 4 and -1

gatecse-2005 linear-algebra eigen-value easy

Answer key

6.3.4 Eigen Value: GATE CSE 2007 | Question: 25



Let A be a 4×4 matrix with eigen values $-5, -2, 1, 4$. Which of the following is an eigen value of the matrix

$\begin{bmatrix} A & I \\ I & A \end{bmatrix}$, where I is the 4×4 identity matrix?

- A. -5 B. -7 C. 2 D. 1

gatecse-2007 eigen-value linear-algebra difficult

Answer key

6.3.5 Eigen Value: GATE CSE 2008 | Question: 28



How many of the following matrices have an eigenvalue 1 ?

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$$

- A. one B. two C. three D. four

gatecse-2008 eigen-value linear-algebra easy

Answer key

6.3.6 Eigen Value: GATE CSE 2010 | Question: 29



Consider the following matrix

$$A = \begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

If the eigenvalues of A are 4 and 8 , then

- A. $x = 4, y = 10$ B. $x = 5, y = 8$ C. $x = 3, y = 9$ D. $x = -4, y = 10$

gatecse-2010 linear-algebra eigen-value easy

Answer key

6.3.7 Eigen Value: GATE CSE 2011 | Question: 40



Consider the matrix as given below.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{bmatrix}$$

Which one of the following options provides the **CORRECT** values of the eigenvalues of the matrix?

- A. 1,4,3 B. 3,7,3 C. 7,3,2 D. 1,2,3

gatecse-2011 linear-algebra eigen-value easy

Answer key

6.3.8 Eigen Value: GATE CSE 2012 | Question: 11



Let A be the 2×2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

- A. 1024 and -1024 B. $1024\sqrt{2}$ and $-1024\sqrt{2}$
C. $4\sqrt{2}$ and $-4\sqrt{2}$ D. $512\sqrt{2}$ and $-512\sqrt{2}$

gatecse-2012 linear-algebra eigen-value easy

Answer key

6.3.9 Eigen Value: GATE CSE 2014 | Set 1 | Question: 5



The value of the dot product of the eigenvectors corresponding to any pair of different eigenvalues of a $4 - by - 4$ symmetric positive definite matrix is _____

gatecse-2014-set1 linear-algebra eigen-value numerical-answers normal

Answer key

6.3.10 Eigen Value: GATE CSE 2014 | Set 2 | Question: 47



The product of the non-zero eigenvalues of the matrix is _____

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

gatecse-2014-set2 linear-algebra eigen-value normal numerical-answers

Answer key

6.3.11 Eigen Value: GATE CSE 2014 | Set 3 | Question: 4



Which one of the following statements is TRUE about every $n \times n$ matrix with only real eigenvalues?

- A. If the trace of the matrix is positive and the determinant of the matrix is negative, at least one of its eigenvalues is negative.
B. If the trace of the matrix is positive, all its eigenvalues are positive.
C. If the determinant of the matrix is positive, all its eigenvalues are positive.
D. If the product of the trace and determinant of the matrix is positive, all its eigenvalues are positive.

gatecse-2014-set3 linear-algebra eigen-value normal

Answer key

6.3.12 Eigen Value: GATE CSE 2015 | Set 1 | Question: 36



Consider the following 2×2 matrix A where two elements are unknown and are marked by a and b . The eigenvalues of this matrix are -1 and 7 . What are the values of a and b ?

$$A = \begin{pmatrix} 1 & 4 \\ b & a \end{pmatrix}$$

- A. $a = 6, b = 4$ B. $a = 4, b = 6$ C. $a = 3, b = 5$ D. $a = 5, b = 3$

gatecse-2015-set1 linear-algebra eigen-value easy

Answer key

6.3.13 Eigen Value: GATE CSE 2015 | Set 2 | Question: 5



The larger of the two eigenvalues of the matrix $\begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$ is _____.

gatecse-2015-set2 linear-algebra eigen-value easy numerical-answers

Answer key

6.3.14 Eigen Value: GATE CSE 2015 | Set 3 | Question: 15



In the given matrix $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, one of the eigenvalues is 1 . The eigenvectors corresponding to the eigenvalue 1 are

- A. $\{a(4, 2, 1) \mid a \neq 0, a \in \mathbb{R}\}$
 B. $\{a(-4, 2, 1) \mid a \neq 0, a \in \mathbb{R}\}$
 C. $\{a(\sqrt{2}, 0, 1) \mid a \neq 0, a \in \mathbb{R}\}$
 D. $\{a(-\sqrt{2}, 0, 1) \mid a \neq 0, a \in \mathbb{R}\}$

gatecse-2015-set3 linear-algebra eigen-value normal

Answer key

6.3.15 Eigen Value: GATE CSE 2016 | Set 1 | Question: 05



Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3 . The determinant of P is _____.

gatecse-2016-set1 linear-algebra eigen-value numerical-answers normal

Answer key

6.3.16 Eigen Value: GATE CSE 2016 | Set 2 | Question: 06



Suppose that the eigenvalues of matrix A are $1, 2, 4$. The determinant of $(A^{-1})^T$ is _____.

gatecse-2016-set2 linear-algebra eigen-value normal numerical-answers

Answer key

6.3.17 Eigen Value: GATE CSE 2017 | Set 1 | Question: 31



Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2 = 50$. Consider the following statements.

- I. One eigenvalue must be in $[-5, 5]$
- II. The eigenvalue with the largest magnitude must be strictly greater than 5

Which of the above statements about eigenvalues of A is/are necessarily CORRECT?

- A. Both I and II B. I only C. II only D. Neither I nor II

Answer key

6.3.18 Eigen Value: GATE CSE 2017 | Set 2 | Question: 52



If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____

gatecse-2017-set2 engineering-mathematics linear-algebra numerical-answers eigen-value

Answer key

6.3.19 Eigen Value: GATE CSE 2018 | Question: 17



Consider a matrix $A = uv^T$ where $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Note that v^T denotes the transpose of v . The largest eigenvalue of A is _____

gatecse-2018 linear-algebra eigen-value normal numerical-answers one-mark

Answer key

6.3.20 Eigen Value: GATE CSE 2018 | Question: 26



Consider a matrix P whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

Consider the following statements.

- I. P does not have an inverse
- II. P has a repeated eigenvalue
- III. P cannot be diagonalized

Which one of the following options is correct?

- A. Only I and III are necessarily true
- B. Only II is necessarily true
- C. Only I and II are necessarily true
- D. Only II and III are necessarily true

gatecse-2018 linear-algebra matrix eigen-value normal two-marks

Answer key

6.3.21 Eigen Value: GATE CSE 2019 | Question: 44



Consider the following matrix:

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

The absolute value of the product of Eigen values of R is _____

gatecse-2019 numerical-answers engineering-mathematics linear-algebra eigen-value two-marks

Answer key

6.3.22 Eigen Value: GATE CSE 2021 | Set 1 | Question: 52



Consider the following matrix.

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

The largest eigenvalue of the above matrix is _____.

gatecse-2021-set1 linear-algebra matrix eigen-value numerical-answers two-marks

Answer key

6.3.23 Eigen Value: GATE CSE 2022 | Question: 43



Which of the following is/are the eigenvector(s) for the matrix given below?

$$\begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$$

A. $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$

B. $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$

C. $\begin{pmatrix} 1 \\ -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$

D. $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$

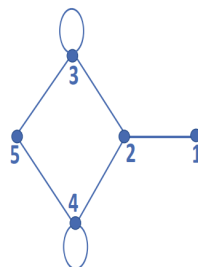
gatecse-2022 linear-algebra eigen-value multiple-selects two-marks

Answer key

6.3.24 Eigen Value: GATE CSE 2023 | Question: 20



Let A be the adjacency matrix of the graph with vertices $\{1, 2, 3, 4, 5\}$.



Let $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, and λ_5 be the five eigenvalues of A . Note that these eigenvalues need not be distinct. The value of $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 =$ _____

gatecse-2023 linear-algebra eigen-value numerical-answers one-mark

Answer key

6.3.25 Eigen Value: GATE CSE 2024 | Set 1 | Question: 2



The product of all eigenvalues of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

A. -1

B. 0

C. 1

D. 2

gatecse-2024-set1 linear-algebra eigen-value one-mark

Answer key

6.3.26 Eigen Value: GATE CSE 2025 | Set 1 | Question: 31



Let A be a 2×2 matrix as given.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

What are the eigenvalues of the matrix A^{13} ?

- A. $1, -1$
 B. $2\sqrt{2}, -2\sqrt{2}$
 C. $4\sqrt{2}, -4\sqrt{2}$
 D. $64\sqrt{2}, -64\sqrt{2}$

gatecse2025-set1 linear-algebra matrix eigen-value easy two-marks

Answer key

6.3.27 Eigen Value: GATE DA 2025 | Question: 18



Let $A = I_n + xx^T$, where I_n is the $n \times n$ identity matrix and $x \in \mathbb{R}^n, x^T x = 1$. Which of the following options is/are correct?

- A. Rank of A is n
 B. A is invertible
 C. 0 is an eigenvalue of A
 D. A^{-1} has a negative eigenvalue

gateda-2025 linear-algebra matrix eigen-value multiple-selects one-mark

Answer key

6.3.28 Eigen Value: GATE DA 2026 | Question: 36



Let $\gamma_1, \gamma_2, \gamma_3$ be the eigenvalues of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \end{bmatrix}$, where $t \in [-\pi, \pi]$ is in radians.

Which one of the following options lists all the possible values of t satisfying $\gamma_1 + \gamma_2 + \gamma_3 = 1 + \sqrt{2}$?

- A. $\left\{ \frac{\pi}{3}, -\frac{\pi}{4} \right\}$
 B. $\left\{ \frac{\pi}{4}, -\frac{\pi}{3} \right\}$
 C. $\left\{ \frac{\pi}{4}, -\frac{\pi}{4} \right\}$
 D. $\left\{ \frac{\pi}{3}, -\frac{\pi}{3} \right\}$

gateda-2026 linear-algebra matrix eigen-value two-marks

Answer key

6.3.29 Eigen Value: GATE DA 2026 | Question: 55



Let $A = (I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T)$ be a matrix, where $\mathbf{1} = (1, 1, 1, \dots, 1)^T \in \mathbb{R}^n$ and I_n is the identity matrix of order n .

The value of $\max_S x^T A x$, where $S = \{x \in \mathbb{R}^n \mid x^T x = 1\}$, is _____. (Answer in integer)

gateda-2026 linear-algebra eigen-value numerical-answers two-marks

Answer key

6.3.30 Eigen Value: GATE DS&AI 2024 | Question: 3



Consider the matrix $M = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$.

Which ONE of the following statements is TRUE?

- A. The eigenvalues of M are non-negative and real.
 B. The eigenvalues of M are complex conjugate pairs.
 C. One eigenvalue of M is positive and real, and another eigenvalue of M is zero.
 D. One eigenvalue of M is non-negative and real, and another eigenvalue of M is negative and real.

gate-ds-ai-2024 eigen-value linear-algebra one-mark

Answer key

6.3.31 Eigen Value: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 17



For matrix $H = \begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$, one of the eigenvalues is 5. Then, the other eigenvalue is

- A. 12 B. 10 C. 8 D. 6

gateda-sample-paper-2024 linear-algebra eigen-value easy

Answer key

6.3.32 Eigen Value: GATE IT 2006 | Question: 26



What are the eigenvalues of the matrix P given below

$$P = \begin{pmatrix} a & 1 & 0 \\ 1 & a & 1 \\ 0 & 1 & a \end{pmatrix}$$

- A. $a, a - \sqrt{2}, a + \sqrt{2}$ B. a, a, a C. $0, a, 2a$ D. $-a, 2a, 2a$

gateit-2006 linear-algebra eigen-value normal

Answer key

6.3.33 Eigen Value: GATE IT 2007 | Question: 2



Let A be the matrix $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. What is the maximum value of $x^T A x$ where the maximum is taken over all x that are the unit eigenvectors of A ?

- A. 5 B. $\frac{(5+\sqrt{5})}{2}$ C. 3 D. $\frac{(5-\sqrt{5})}{2}$

gateit-2007 linear-algebra eigen-value normal

Answer key

6.4 Gaussian Elimination (1)

6.4.1 Gaussian Elimination: GATE DA 2025 | Question: 2



The number of additions and multiplications involved in performing Gaussian elimination on any $n \times n$ upper triangular matrix is of the order

- A. $O(n)$ B. $O(n^2)$ C. $O(n^3)$ D. $O(n^4)$

gateda-2025 linear-algebra matrix gaussian-elimination one-mark

Answer key

6.5 Lu Decomposition (3)

6.5.1 Lu Decomposition: GATE CSE 2015 | Set 1 | Question: 18



In the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 \\ 4 & 9 \end{bmatrix}$, if the diagonal elements of U are both 1, then the lower diagonal entry l_{22} of L is_____.

gatecse-2015-set1 linear-algebra lu-decomposition numerical-answers

Answer key

6.5.2 Lu Decomposition: GATE CSE 2022 | Question: 35



Consider solving the following system of simultaneous equations using LU decomposition.

$$x_1 + x_2 - 2x_3 = 4$$

$$x_1 + 3x_2 - x_3 = 7$$

$$2x_1 + x_2 - 5x_3 = 7$$

where L and U are denoted as

$$L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}$$

Which one of the following is the correct combination of values for L_{32} , U_{33} , and x_1 ?

A. $L_{32} = 2, U_{33} = -\frac{1}{2}, x_1 = -1$

B. $L_{32} = 2, U_{33} = 2, x_1 = -1$

C. $L_{32} = -\frac{1}{2}, U_{33} = 2, x_1 = 0$

D. $L_{32} = -\frac{1}{2}, U_{33} = -\frac{1}{2}, x_1 = 0$

gatecse-2022 linear-algebra lu-decomposition matrix system-of-equations two-marks

Answer key

6.5.3 Lu Decomposition: GATE CSE 2025 | Set 2 | Question: 34



Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3 \times 3}$ and $Q \in \mathbb{R}^{3 \times 1}$. Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

Which of the following statement(s) is/are TRUE?

- A. The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
- B. If P is invertible, then both L and U are invertible.
- C. If P is singular, then at least one of the diagonal elements of U is zero.
- D. If P is symmetric, then both L and U are symmetric.

gatecse2025-set2 linear-algebra system-of-equations matrix lu-decomposition multiple-selects two-marks

Answer key

6.6

Matrix (24)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#)

6.6.1 Matrix: GATE CSE 1987 | Question: 1-xxiii



A square matrix is singular whenever

- A. The rows are linearly independent
- B. The columns are linearly independent
- C. The row are linearly dependent
- D. None of the above

gate1987 linear-algebra matrix

Answer key

6.6.2 Matrix: GATE CSE 1988 | Question: 16i



Assume that the matrix A given below, has factorization of the form $LU = PA$, where L is lower-triangular with all diagonal elements equal to 1, U is upper-triangular, and P is a permutation matrix. For

$$A = \begin{bmatrix} 2 & 5 & 9 \\ 4 & 6 & 5 \\ 8 & 2 & 3 \end{bmatrix}$$

Compute L , U , and P using Gaussian elimination with partial pivoting.

gate1988 normal descriptive linear-algebra matrix

Answer key 

6.6.3 Matrix: GATE CSE 1993 | Question: 02.7



If $A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & i & i \\ 0 & 0 & 0 & -i \end{pmatrix}$ the matrix A^4 , calculated by the use of Cayley-Hamilton theorem or otherwise, is _____

gate1993 linear-algebra normal matrix fill-in-the-blanks

Answer key 

6.6.4 Matrix: GATE CSE 1994 | Question: 1.2



Let A and B be real symmetric matrices of size $n \times n$. Then which one of the following is true?

- A. $AA' = I$ B. $A = A^{-1}$ C. $AB = BA$ D. $(AB)' = BA$

gate1994 linear-algebra normal matrix

Answer key 

6.6.5 Matrix: GATE CSE 1994 | Question: 3.12



Find the inverse of the matrix $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

gate1994 linear-algebra matrix easy descriptive

Answer key 

6.6.6 Matrix: GATE CSE 1996 | Question: 10



Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ be two matrices such that $AB = I$. Let $C = A \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $CD = I$. Express the elements of D in terms of the elements of B .

gate1996 linear-algebra matrix normal descriptive

Answer key 

6.6.7 Matrix: GATE CSE 1996 | Question: 2.6



The matrices $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ and $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under multiplication

- A. if $a = b$ or $\theta = n\pi, n$ an integer B. always
C. never D. if $a \cos \theta = b \sin \theta$

gate1996 linear-algebra normal matrix

Answer key

6.6.8 Matrix: GATE CSE 1997 | Question: 4.2



Let $A = (a_{ij})$ be an n -rowed square matrix and I_{12} be the matrix obtained by interchanging the first and second rows of the n -rowed Identity matrix. Then AI_{12} is such that its first

- A. Row is the same as its second row
- B. Row is the same as the second row of A
- C. Column is the same as the second column of A
- D. Row is all zero

gate1997 linear-algebra easy matrix

Answer key

6.6.9 Matrix: GATE CSE 1998 | Question: 2.2



Consider the following determinant $\Delta = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$

Which of the following is a factor of Δ ?

- A. $a + b$
- B. $a - b$
- C. $a + b + c$
- D. abc

gate1998 linear-algebra matrix normal

Answer key

6.6.10 Matrix: GATE CSE 2001 | Question: 1.1



Consider the following statements:

- S1: The sum of two singular $n \times n$ matrices may be non-singular
- S2: The sum of two $n \times n$ non-singular matrices may be singular

Which one of the following statements is correct?

- A. $S1$ and $S2$ both are true
- B. $S1$ is true, $S2$ is false
- C. $S1$ is false, $S2$ is true
- D. $S1$ and $S2$ both are false

gatecse-2001 linear-algebra normal matrix

Answer key

6.6.11 Matrix: GATE CSE 2004 | Question: 26



The number of different $n \times n$ symmetric matrices with each element being either 0 or 1 is: (Note: power $(2, X)$ is same as 2^X)

- A. power $(2, n)$
- B. power $(2, n^2)$
- C. power $\left(2, \frac{n^2+n}{2}\right)$
- D. power $\left(2, \frac{n^2-n}{2}\right)$

gatecse-2004 linear-algebra normal matrix

Answer key

6.6.12 Matrix: GATE CSE 2004 | Question: 27



Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = I$, then B^{-1} is

- A. $D^{-1}C^{-1}A^{-1}$
- B. CDA
- C. ADC
- D. Does not necessarily exist

gatecse-2004 linear-algebra normal matrix

Answer key

6.6.13 Matrix: GATE CSE 2004 | Question: 76



In an $M \times N$ matrix all non-zero entries are covered in a rows and b columns. Then the maximum number of non-zero entries, such that no two are on the same row or column, is

- A. $\leq a + b$
 B. $\leq \max(a, b)$
 C. $\leq \min(M - a, N - b)$
 D. $\leq \min(a, b)$

gatecse-2004 linear-algebra normal matrix

Answer key

6.6.14 Matrix: GATE CSE 2006 | Question: 23



F is an $n \times n$ real matrix. b is an $n \times 1$ real vector. Suppose there are two $n \times 1$ vectors, u and v such that, $u \neq v$ and $Fu = b, Fv = b$. Which one of the following statements is false?

- A. Determinant of F is zero.
 B. There are an infinite number of solutions to $Fx = b$
 C. There is an $x \neq 0$ such that $Fx = 0$
 D. F must have two identical rows

gatecse-2006 linear-algebra normal matrix

Answer key

6.6.15 Matrix: GATE CSE 2015 | Set 2 | Question: 27



Perform the following operations on the matrix $\begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$

- i. Add the third row to the second row
 ii. Subtract the third column from the first column.

The determinant of the resultant matrix is _____.

gatecse-2015-set2 linear-algebra matrix easy numerical-answers

Answer key

6.6.16 Matrix: GATE CSE 2022 | Question: 10



Consider the following two statements with respect to the matrices $A_{m \times n}, B_{n \times m}, C_{n \times n}$ and $D_{n \times n}$.

Statement 1 : $tr(AB) = tr(BA)$

Statement 2 : $tr(CD) = tr(DC)$

where $tr()$ represents the trace of a matrix. Which one of the following holds?

- A. Statement 1 is correct and Statement 2 is wrong.
 B. Statement 1 is wrong and Statement 2 is correct.
 C. Both Statement 1 and Statement 2 are correct.
 D. Both Statement 1 and Statement 2 are wrong.

gatecse-2022 linear-algebra matrix one-mark

Answer key

6.6.17 Matrix: GATE CSE 2025 | Set 2 | Question: 1



If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, then which ONE of the following is A^8 ?

- A. $\begin{pmatrix} 25 & 0 \\ 0 & 25 \end{pmatrix}$
 B. $\begin{pmatrix} 125 & 0 \\ 0 & 125 \end{pmatrix}$

C. $\begin{pmatrix} 625 & 0 \\ 0 & 625 \end{pmatrix}$

gatecse2025-set2 linear-algebra matrix easy one-mark

Answer key

D. $\begin{pmatrix} 3125 & 0 \\ 0 & 3125 \end{pmatrix}$

6.6.18 Matrix: GATE CSE 2026 | Set 1 | Question: 10



Let $n > 1$. Consider an $n \times n$ matrix M with its elements from $\mathbb{R}$. Let the vector $(0, 1, 0, 0, \dots, 0) \in \mathbb{R}^n$ be in the null space of M .

Which of the following options is/are always correct?

- A. Determinant of M is 1
- B. Determinant of M is 0
- C. Rank of M is 1
- D. There are at least two non-zero vectors in the null space of M

gatecse-2026-set1 linear-algebra matrix multiple-selects one-mark

Answer key

6.6.19 Matrix: GATE DA 2025 | Question: 27



Let $A \in \mathbb{R}^{n \times n}$ be such that $A^3 = A$. Which one of the following statements is ALWAYS correct?

- A. A is invertible
- B. Determinant of A is 0
- C. The sum of the diagonal elements of A is 1
- D. A and A^2 have the same rank

gateda-2025 linear-algebra matrix two-marks

Answer key

6.6.20 Matrix: GATE DA 2025 | Question: 42



An $n \times n$ matrix A with real entries satisfies the property: $\|Ax\|^2 = \|x\|^2$, for all $x \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm. Which of the following statements is/are ALWAYS correct?

- A. A must be orthogonal
- B. $A = I$, where I denotes the identity matrix, is the only solution
- C. The eigenvalues of A are either $+1$ or -1
- D. A has full rank

gateda-2025 linear-algebra matrix multiple-selects two-marks

Answer key

6.6.21 Matrix: GATE DA 2026 | Question: 11



Let $M = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ be a 2×2 matrix, where $\theta = \frac{2\pi}{5}$, and $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Which of the following options is equal to M^{2026} ?

- A. M^2
- B. M
- C. M^{-1}
- D. I_2

gateda-2026 linear-algebra matrix one-mark

Answer key

6.6.22 Matrix: GATE DA 2026 | Question: 42



Let $M = (I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T)$ be a matrix, where $\mathbf{1} = (1, 1, 1, \dots, 1)^T \in \mathbb{R}^n$ and I_n is the identity matrix of order n .

Which of the following options is/are correct?

- A. $M^T = M$
- B. $M^2 = I_n$
- C. $\text{Trace}(M) = n$
- D. M is a projection matrix

Answer key

6.6.23 Matrix: GATE IT 2004 | Question: 36



If matrix $X = \begin{bmatrix} a & 1 \\ -a^2 + a - 1 & 1 - a \end{bmatrix}$ and $X^2 - X + I = O$ (I is the identity matrix and O is the zero matrix), then the inverse of X is

A. $\begin{bmatrix} 1-a & -1 \\ a^2 & a \end{bmatrix}$

B. $\begin{bmatrix} 1-a & -1 \\ a^2 - a + 1 & a \end{bmatrix}$

C. $\begin{bmatrix} -a & 1 \\ -a^2 + a - 1 & 1 - a \end{bmatrix}$

D. $\begin{bmatrix} a^2 - a + 1 & a \\ 1 & 1 - a \end{bmatrix}$

gateit-2004 linear-algebra matrix normal

Answer key

6.6.24 Matrix: GATE IT 2008 | Question: 29



If M is a square matrix with a zero determinant, which of the following assertion (s) is (are) correct?

S1: Each row of M can be represented as a linear combination of the other rows

S2: Each column of M can be represented as a linear combination of the other columns

S3: $MX = 0$ has a nontrivial solution

S4: M has an inverse

A. $S3$ and $S2$

B. $S1$ and $S4$

C. $S1$ and $S3$

D. $S1, S2$ and $S3$

gateit-2008 linear-algebra normal matrix

Answer key

6.7 Multiplicity (1)

6.7.1 Multiplicity: GATE CSE 2026 | Set 1 | Question: 3



For $n > 1$, the maximum multiplicity of any eigenvalue of an $n \times n$ matrix with elements from $\mathbb{R}$ is

A. n

B. $n - 1$

C. 1

D. $n + 1$

gatecse-2026-set1 eigen-value multiplicity linear-algebra one-mark

Answer key

6.8 Orthonormality (2)

6.8.1 Orthonormality: GATE DA 2025 | Question: 15



Which of the following statements is/are correct?

A. $\mathbb{R}^n$ has a unique set of orthonormal basis vectors

B. $\mathbb{R}^n$ does not have a unique set of orthonormal basis vectors

C. Linearly independent vectors in $\mathbb{R}^n$ are orthonormal

D. Orthonormal vectors in $\mathbb{R}^n$ are linearly independent

gateda-2025 linear-algebra vector-space orthonormality multiple-selects one-mark

Answer key

6.8.2 Orthonormality: GATE DA 2025 | Question: 40



Let x_1, x_2, x_3, x_4, x_5 be a system of orthonormal vectors in $\mathbb{R}^{10}$. Consider the matrix $A = x_1 x_1^\top + \dots + x_5 x_5^\top$. Which of the following statements is/are correct?

A. Singular values of A are also its eigenvalues

B. Singular values of A are either 0 or 1

C. Determinant of A is 1

D. A is invertible

gateda-2025 linear-algebra orthonormality vector-space matrix multiple-selects two-marks

Answer key

6.9

Rank of Matrix (7)

Practice Tests: Test 1 (15Q) Test 2 (6Q)

6.9.1 Rank of Matrix: GATE CSE 1994 | Question: 1.9



The rank of matrix $\begin{bmatrix} 0 & 0 & -3 \\ 9 & 3 & 5 \\ 3 & 1 & 1 \end{bmatrix}$ is:

A. 0

B. 1

C. 2

D. 3

gate1994 linear-algebra matrix rank-of-matrix easy

Answer key

6.9.2 Rank of Matrix: GATE CSE 1995 | Question: 1.24



The rank of the following $(n+1) \times (n+1)$ matrix, where a is a real number is

$$\begin{bmatrix} 1 & a & a^2 & \dots & a^n \\ 1 & a & a^2 & \dots & a^n \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a & a^2 & \dots & a^n \end{bmatrix}$$

A. 1

C. n

B. 2

D. Depends on the value of a

gate1995 linear-algebra matrix normal rank-of-matrix

Answer key

6.9.3 Rank of Matrix: GATE CSE 1998 | Question: 2.1



The rank of the matrix given below is:

$$\begin{bmatrix} 1 & 4 & 8 & 7 \\ 0 & 0 & 3 & 0 \\ 4 & 2 & 3 & 1 \\ 3 & 12 & 24 & 21 \end{bmatrix}$$

A. 3

B. 1

C. 2

D. 4

gate1998 linear-algebra matrix normal rank-of-matrix

Answer key

6.9.4 Rank of Matrix: GATE CSE 2002 | Question: 1.1



The rank of the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ is

A. 4

B. 2

C. 1

D. 0

gatecse-2002 linear-algebra easy rank-of-matrix

Answer key

6.9.5 Rank of Matrix: GATE CSE 2017 | Set 2 | Question: 22



Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices.

Then the rank of $P + Q$ is _____.

gatecse-2017-set2 linear-algebra rank-of-matrix numerical-answers

Answer key

6.9.6 Rank of Matrix: GATE CSE 2020 | Question: 27



Let A and B be two $n \times n$ matrices over real numbers. Let $\text{rank}(M)$ and $\det(M)$ denote the rank and determinant of a matrix M , respectively. Consider the following statements.

- I. $\text{rank}(AB) = \text{rank}(A)\text{rank}(B)$
- II. $\det(AB) = \det(A)\det(B)$
- III. $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$
- IV. $\det(A+B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

- A. I and II only B. I and IV only C. II and III only D. III and IV only

gatecse-2020 linear-algebra matrix two-marks rank-of-matrix

Answer key

6.9.7 Rank of Matrix: GATE CSE 2021 | Set 2 | Question: 24



Suppose that P is a 4×5 matrix such that every solution of the equation $Px=0$ is a scalar multiple of $\begin{bmatrix} 2 & 5 & 4 & 3 & 1 \end{bmatrix}^T$. The rank of P is _____.

gatecse-2021-set2 numerical-answers linear-algebra matrix rank-of-matrix one-mark

Answer key

6.10

Singular Value Decomposition (1)

Practice Test: Test 1 (5Q)

6.10.1 Singular Value Decomposition: GATE DS&AI 2024 | Question: 51



Let $u = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$, and let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5$ be the singular values of the matrix $M = uu^T$ (where u^T is the transpose of u). The value of $\sum_{i=1}^5 \sigma_i$ is _____.

gate-ds-ai-2024 linear-algebra singular-value-decomposition numerical-answers two-marks

Answer key

6.11

Statistics (2)

Practice Test: Test 1 (12Q)

6.11.1 Statistics: GATE 2023 Stats Exam Q. 13



Consider the following statements.

- (I) Let A and B be two $n \times n$ real matrices. If B is invertible, then $\text{rank}(BA) = \text{rank}(A)$.

(II) Let A be an $n \times n$ real matrix. If $A^2 \mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^n$, then $A \mathbf{x} = \mathbf{b}$ also has a solution for every $\mathbf{b} \in \mathbb{R}^n$.

Which of the above statements is/are true?

- A. & Only (I) B. & Only (II)
 C. & Both (I) and (II) D. & Neither (I) nor (II)

gate-preparation linear-algebra statistics

Answer key

6.11.2 Statistics: GATE STATISTICS 2021 Q.13



Let A be the 2×2 real matrix having eigenvalues 1 and -1 , with corresponding eigenvectors $\begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix}$ and $\begin{bmatrix} \frac{-1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$, respectively. If $A^{2021} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $a + b + c + d$ equals ____ (round off to 2 decimal places).

statistics linear-algebra gate-preparation

Answer key

6.12

Subspace (1)

6.12.1 Subspace: GATE DS&AI 2024 | Question: 37



Select all choices that are subspaces of $\mathbb{R}^3$.

Note: $\mathbb{R}$ denotes the set of real numbers.

- A. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
- B. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \alpha, \beta \in \mathbb{R} \right\}$
- C. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 = 0, 4x_1 - 2x_2 + 3x_3 = 0 \right\}$
- D. $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 + 4 = 0 \right\}$

gate-ds-ai-2024 linear-algebra vector-space subspace multiple-selects two-marks

Answer key

6.13

System of Equations (17)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(7Q\)](#)

6.13.1 System of Equations: GATE CSE 1996 | Question: 1.7



Let $Ax = b$ be a system of linear equations where A is an $m \times n$ matrix and b is a $m \times 1$ column vector and X is an $n \times 1$ column vector of unknowns. Which of the following is false?

- A. The system has a solution if and only if, both A and the augmented matrix $[Ab]$ have the same rank.
- B. If $m < n$ and b is the zero vector, then the system has infinitely many solutions.
- C. If $m = n$ and b is a non-zero vector, then the system has a unique solution.
- D. The system will have only a trivial solution when $m = n$, b is the zero vector and $\text{rank}(A) = n$.

gate1996 linear-algebra system-of-equations normal

Answer key

6.13.2 System of Equations: GATE CSE 1998 | Question: 1.2



Consider the following set of equations

- $x + 2y = 5$
- $4x + 8y = 12$
- $3x + 6y + 3z = 15$

This set

- A. has unique solution
- B. has no solution
- C. has finite number of solutions
- D. has infinite number of solutions

gate1998 linear-algebra system-of-equations easy

Answer key

6.13.3 System of Equations: GATE CSE 1998 | Question: 9



Derive the expressions for the number of operations required to solve a system of linear equations in n unknowns using the Gaussian Elimination Method. Assume that one operation refers to a multiplication followed by an addition.

gate1998 linear-algebra system-of-equations descriptive

Answer key

6.13.4 System of Equations: GATE CSE 2003 | Question: 41



Consider the following system of linear equations

$$\begin{pmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha \\ 5 \\ 7 \end{pmatrix}$$

Notice that the second and the third columns of the coefficient matrix are linearly dependent. For how many values of α , does this system of equations have infinitely many solutions?

- A. 0
- B. 1
- C. 2
- D. 3

gatecse-2003 linear-algebra system-of-equations normal

Answer key

6.13.5 System of Equations: GATE CSE 2004 | Question: 71



How many solutions does the following system of linear equations have?

- $-x + 5y = -1$
- $x - y = 2$
- $x + 3y = 3$

- A. infinitely many
- B. two distinct solutions

C. unique

D. none

gatecse-2004 linear-algebra system-of-equations normal

Answer key

6.13.6 System of Equations: GATE CSE 2005 | Question: 48



Consider the following system of linear equations :

$$2x_1 - x_2 + 3x_3 = 1$$

$$3x_1 + 2x_2 + 5x_3 = 2$$

$$-x_1 + 4x_2 + x_3 = 3$$

The system of equations has

A. no solution

B. a unique solution

C. more than one but a finite number of solutions

D. an infinite number of solutions

gatecse-2005 linear-algebra system-of-equations normal

Answer key

6.13.7 System of Equations: GATE CSE 2008 | Question: 3



The following system of equations

- $x_1 + x_2 + 2x_3 = 1$
- $x_1 + 2x_2 + 3x_3 = 2$
- $x_1 + 4x_2 + \alpha x_3 = 4$

has a unique solution. The only possible value(s) for α is/are

A. 0

B. either 0 or 1

C. one of 0, 1, or -1

D. any real number

gatecse-2008 easy linear-algebra system-of-equations

Answer key

6.13.8 System of Equations: GATE CSE 2014 | Set 1 | Question: 4



Consider the following system of equations:

- $3x + 2y = 1$
- $4x + 7z = 1$
- $x + y + z = 3$
- $x - 2y + 7z = 0$

The number of solutions for this system is _____

gatecse-2014-set1 linear-algebra system-of-equations numerical-answers normal

Answer key

6.13.9 System of Equations: GATE CSE 2015 | Set 3 | Question: 33



If the following system has non-trivial solution,

- $px + qy + rz = 0$
- $qx + ry + pz = 0$
- $rx + py + qz = 0,$

then which one of the following options is TRUE?

A. $p - q + r = 0$ or $p = q = -r$

B. $p + q - r = 0$ or $p = -q = r$

C. $p + q + r = 0$ or $p = q = r$

D. $p - q + r = 0$ or $p = -q = -r$

gatecse-2015-set3 linear-algebra system-of-equations normal

6.13.10 System of Equations: GATE CSE 2016 | Set 2 | Question: 04



Consider the systems, each consisting of m linear equations in n variables.

- I. If $m < n$, then all such systems have a solution.
- II. If $m > n$, then none of these systems has a solution.
- III. If $m = n$, then there exists a system which has a solution.

Which one of the following is **CORRECT**?

- A. I, II and III are true.
- B. Only II and III are true.
- C. Only III is true.
- D. None of them is true.

gatecse-2016-set2 linear-algebra system-of-equations normal

6.13.11 System of Equations: GATE CSE 2017 | Set 1 | Question: 3



Let $c_1, \dots, c_n$ be scalars, not all zero, such that $\sum_{i=1}^n c_i a_i = 0$ where a_i are column vectors in R^n .

Consider the set of linear equations

$$Ax = b$$

where $A = [a_1 \dots a_n]$ and $b = \sum_{i=1}^n a_i$. The set of equations has

- A. a unique solution at $x = J_n$ where J_n denotes a n -dimensional vector of all 1.
- B. no solution
- C. infinitely many solutions
- D. finitely many solutions

gatecse-2017-set1 linear-algebra system-of-equations normal

6.13.12 System of Equations: GATE CSE 2024 | Set 1 | Question: 39



Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

- A. There exist at least $m - n$ linearly independent solutions to this system
- B. There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors
- C. There exists a non-zero solution in which at least $m - n$ variables are 0
- D. There exists a solution in which at least n variables are non-zero

gatecse-2024-set1 multiple-selects linear-algebra system-of-equations two-marks

6.13.13 System of Equations: GATE CSE 2025 | Set 1 | Question: 13



Consider the given system of linear equations for variables x and y , where k is a real-valued constant. Which of the following option(s) is/are CORRECT?

$$\begin{aligned} x + ky &= 1 \\ kx + y &= -1 \end{aligned}$$

- A. There is exactly one value of k for which the above system of equations has no solution.
- B. There exist an infinite number of values of k for which the system of equations has no solution.
- C. There exists exactly one value of k for which the system of equations has exactly one solution.
- D. There exists exactly one value of k for which the system of equations has an infinite number of solutions.

gatecse2025-set1 linear-algebra system-of-equations multiple-selects one-mark

Answer key

6.13.14 System of Equations: GATE CSE 2026 | Set 2 | Question: 21



Consider the system of linear equations given below.

$$\begin{aligned} ax + y &= b \\ 16x + ay &= 24 \end{aligned}$$

Suppose the values of a and b are chosen such that the system of linear equations produce multiple solutions. Then the product of a and b is _____. (answer in integer)

gatecse-2026-set2 linear-algebra system-of-equations numerical-answers one-mark

Answer key

6.13.15 System of Equations: GATE DA 2025 | Question: 3



The sum of the elements in each row of $A \in \mathbb{R}^{n \times n}$ is 1. If $B = A^3 - 2A^2 + A$, which one of the following statements is correct (for $x \in \mathbb{R}^n$)?

- A. The equation $Bx = 0$ has no solution
- B. The equation $Bx = 0$ has exactly two solutions
- C. The equation $Bx = 0$ has infinitely many solutions
- D. The equation $Bx = 0$ has a unique solution

gateda-2025 linear-algebra system-of-equations one-mark

Answer key

6.13.16 System of Equations: GATE DS&AI 2024 | Question: 38



Which of the following statements is/are TRUE?

Note: $\mathbb{R}$ denotes the set of real numbers.

- A. There exist $M \in \mathbb{R}^{3 \times 3}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has a unique solution and $Mx = q$ has infinite solutions.
- B. There exist $M \in \mathbb{R}^{3 \times 3}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has no solutions and $Mx = q$ has infinite solutions.
- C. There exist $M \in \mathbb{R}^{2 \times 3}$, $p \in \mathbb{R}^2$, and $q \in \mathbb{R}^2$ such that $Mx = p$ has a unique solution and $Mx = q$ has infinite solutions.
- D. There exist $M \in \mathbb{R}^{3 \times 2}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has a unique solution and $Mx = q$ has no solutions.

linear-algebra system-of-equations gate-ds-ai-2024 multiple-selects two-marks

Answer key

6.13.17 System of Equations: GATE IT 2004 | Question: 6



What values of x , y and z satisfy the following system of linear equations?

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 12 \end{bmatrix}$$

- A. $x = 6, y = 3, z = 2$
- B. $x = 12, y = 3, z = -4$
- C. $x = 6, y = 6, z = -4$
- D. $x = 12, y = -3, z = 0$

gateit-2004 linear-algebra system-of-equations easy

Answer key

 **Practice Tests:** [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(5Q\)](#)

6.14.1 Vector Space: GATE CSE 1995 | Question: 2.13



A unit vector perpendicular to both the vectors $a = 2i - 3j + k$ and $b = i + j - 2k$ is:

- A. $\frac{1}{\sqrt{3}}(i + j + k)$ B. $\frac{1}{3}(i + j - k)$ C. $\frac{1}{3}(i - j - k)$ D. $\frac{1}{\sqrt{3}}(i + j - k)$

gate1995 linear-algebra normal vector-space

[Answer key](#)

6.14.2 Vector Space: GATE CSE 2007 | Question: 27



Consider the set of (column) vectors defined by

$$X = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0, \text{ where } x^T = [x_1, x_2, x_3]^T\}$$

.Which of the following is TRUE?

- A. $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a basis for the subspace X .
 B. $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X .
 C. X is not a subspace of $\mathbb{R}^3$.
 D. None of the above

gatecse-2007 linear-algebra normal vector-space

[Answer key](#)

6.14.3 Vector Space: GATE CSE 2014 | Set 3 | Question: 5



If V_1 and V_2 are 4-dimensional subspaces of a 6-dimensional vector space V , then the smallest possible dimension of $V_1 \cap V_2$ is ____.

gatecse-2014-set3 linear-algebra vector-space normal numerical-answers

[Answer key](#)

6.14.4 Vector Space: GATE CSE 2017 | Set 1 | Question: 30



Let u and v be two vectors in $\mathbb{R}^2$ whose Euclidean norms satisfy $\|u\| = 2\|v\|$. What is the value of α such that $w = u + \alpha v$ bisects the angle between u and v ?

- A. 2 B. $\frac{1}{2}$ C. 1 D. $-\frac{1}{2}$

gatecse-2017-set1 linear-algebra normal vector-space

[Answer key](#)

6.14.5 Vector Space: GATE DA 2025 | Question: 28



Let $\{x_1, x_2, \dots, x_n\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. Let the (i, j) -th element of matrix $A \in \mathbb{R}^{n \times n}$ be given by $A_{ij} = x_i^T x_j, 1 \leq i, j \leq n$. Which one of the following statements is correct?

- A. A is invertible B. 0 is a singular value of A
 C. Determinant of A is 0 D. $z^T A z = 0$ for some non-zero $z \in \mathbb{R}^n$

gateda-2025 linear-algebra vector-space matrix two-marks

[Answer key](#)

6.14.6 Vector Space: GATE DA 2026 | Question: 12



Consider a set $S_1 = \{x = (x_1, x_2, x_3)^T \in \mathbb{R}^3 \mid x^T x \leq 16\}$. Let S_2 be another set which is a subspace of $\mathbb{R}^3$ with dimension two.

Which of the following gives the area of $S_1 \cap S_2$?

- A. 16π B. 4π C. $4\pi^2$ D. $16\pi^2$

gateda-2026 linear-algebra vector-space one-mark

Answer key

6.14.7 Vector Space: GATE DS&AI 2024 | Question: 39



Let $\mathbb{R}$ be the set of real numbers, U be a subspace of $\mathbb{R}^3$ and $M \in \mathbb{R}^{3 \times 3}$ be the matrix corresponding to the projection on to the subspace U .

Which of the following statements is/are TRUE?

- A. If U is a 1-dimensional subspace of $\mathbb{R}^3$, then the null space of M is a 1-dimensional subspace.
 B. If U is a 2-dimensional subspace of $\mathbb{R}^3$, then the null space of M is a 1-dimensional subspace.
 C. $M^2 = M$
 D. $M^3 = M$

gate-ds-ai-2024 linear-algebra vector-space multiple-selects two-marks

Answer key

Answer Keys

6.1.1	A	6.2.1	B	6.2.2	A	6.2.3	A	6.2.4	0
6.2.5	D	6.2.6	B	6.2.7	A;B	6.2.8	A	6.2.9	48:48
6.2.10	0	6.2.11	D	6.2.12	A	6.3.1	B;D	6.3.2	N/A
6.3.3	B	6.3.4	C	6.3.5	A	6.3.6	D	6.3.7	A
6.3.8	D	6.3.9	0	6.3.10	6	6.3.11	A	6.3.12	D
6.3.13	6	6.3.14	B	6.3.15	15	6.3.16	0.125	6.3.17	B
6.3.18	5	6.3.19	3	6.3.20	D	6.3.21	12	6.3.22	3 : 3
6.3.23	A;C;D	6.3.24	2	6.3.25	B	6.3.26	D	6.3.27	A;B
6.3.28	C	6.3.29	1:1	6.3.30	B	6.3.31	B	6.3.32	A
6.3.33	B	6.4.1	B	6.5.1	5	6.5.2	D	6.5.3	A;B;C
6.6.1	C	6.6.2	N/A	6.6.3	N/A	6.6.4	D	6.6.5	N/A
6.6.6	N/A	6.6.7	A	6.6.8	C	6.6.9	B	6.6.10	A
6.6.11	C	6.6.12	B	6.6.13	D	6.6.14	D	6.6.15	0
6.6.16	C	6.6.17	C	6.6.18	B;D	6.6.19	D	6.6.20	A;D
6.6.21	B	6.6.22	A;D	6.6.23	B	6.6.24	D	6.7.1	A
6.8.1	B;D	6.8.2	A;B	6.9.1	C	6.9.2	A	6.9.3	A
6.9.4	C	6.9.5	2	6.9.6	C	6.9.7	4 : 4	6.10.1	55
6.11.1	TBA	6.11.2	TBA	6.12.1	A;C	6.13.1	C	6.13.2	B
6.13.3	N/A	6.13.4	B	6.13.5	C	6.13.6	B	6.13.7	X

6.13.8	1
6.13.13	A;D
6.14.1	A
6.14.6	A

6.13.9	C
6.13.14	24:24
6.14.2	A
6.14.7	B;C;D

6.13.10	C
6.13.15	C
6.14.3	2

6.13.11	C
6.13.16	B;D
6.14.4	A

6.13.12	A
6.13.17	C
6.14.5	A